

Task 2 — AI Generated Snake Game Report

Project Title

Snake Survival Arena

Deployed Game/Application Link

<https://aisha-batool.github.io/ai-prompting-asg1/>

AI Tool Used

For this project, I used **ChatGPT** as the AI coding assistant. ChatGPT was used to generate the initial game code, improve the game mechanics, fix bugs, update the user interface, and create a more polished final version of the game.

Technology Stack

The game was created as a web-based application using:

- HTML
- CSS
- JavaScript
- HTML Canvas
- GitHub Pages for deployment

No backend or database was required because the game runs directly inside the browser.

1. Initial Game Idea

The original idea was to create a Snake Game with more advanced mechanics than the traditional version. Instead of only eating food, the player snake could also interact with other snakes inside the arena.

The main idea was:

- The player controls a snake.
- The snake wraps around the screen instead of hitting walls.

- The game difficulty increases as the player continues playing.
- The player can eat smaller snakes.
- Bigger snakes and poisonous snakes are dangerous.
- If the player hits or is hit by a dangerous snake, the game ends.
- The game should be web-based and deployable on GitHub Pages.

This idea was given to ChatGPT as the first prompt.

2. Prompt Iterations Used During Development

Prompt 1 — Initial Game Generation

Prompt used:

I want to make a snake game. I want it to have increasing level and difficulty as the player continues to play. Snake will wrap around the screen. It can eat other snakes as well if the snake is small. But some snakes which are big or harmful, if it eats them, it will die. I want the game to be web based so I want to sync it on GitHub.

Result:

ChatGPT generated the first version of the game using HTML, CSS, and JavaScript. It included a basic snake, food, enemy snakes, score, level system, and wrap-around movement.

What worked:

- The basic structure was created.
- The game could run in a browser.
- The player snake could move.
- Score and level were included.
- GitHub Pages deployment was possible.

What did not work well:

- Enemy snakes were too simple.
- The game did not feel polished.
- Some game rules were not clear.

- The visual design was basic.
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Prompt 2 — GitHub Deployment

Prompt used:

Can we not push it directly from ChatGPT? Sync etc.

Result:

ChatGPT explained that direct pushing to GitHub was not possible without GitHub authentication. It suggested creating a zip file and uploading the files manually to GitHub.

What worked:

- I understood the GitHub workflow.
- The project files were packaged into a zip file.
- I uploaded the files to GitHub and published the project using GitHub Pages.

What did not work well:

- Direct GitHub sync was not possible from ChatGPT.
 - The files had to be uploaded manually.
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Prompt 3 — Fixing the Game Logic

Prompt used:

The game is not working correctly. Think hard.

Result:

The game was rebuilt with better logic. The new version included a proper start screen, pause button, restart button, better enemy snakes, safer spawning, high score, and mobile controls.

What worked:

- The game became more stable.
- Enemy snakes had body lengths.
- Small, big, and poison snakes became more meaningful.

- The game had better user controls.

What did not work well:

- The visual appearance still needed improvement.
 - The landing page was still not like a professional game website.
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Prompt 4 — Landing Page and Realistic Snake Design

Prompt used:

I want a good landing page. Also snakes should look like a real snake.

Result:

Initially, ChatGPT generated an image mockup of a professional landing page. However, this was only an image and not actual code.

What worked:

- The image gave a clear design direction.
- It showed a premium dark jungle-style landing page.
- It helped define the final visual style of the project.

What did not work:

- The image itself could not be used as a working website.
 - The actual code still needed to be updated.
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Prompt 5 — Converting the Mockup into Code

Prompt used:

This is just the image. I want to update my game code.

Result:

ChatGPT updated the actual code and created a better landing page with sections such as hero area, features, gameplay preview, how-to-play section, and playable game section.

What worked:

- The game now had a proper website structure.
- The snake visuals were improved using canvas drawing.
- The player and enemy snakes had curved bodies, eyes, tongue, and scale-like highlights.

What did not work well:

- The landing page still needed to match the earlier image more closely.
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Prompt 6 — Matching the Earlier Landing Page Image

Prompt used:

I want the landing page UI to be the image you shared earlier.

Result:

The landing page was redesigned to closely match the earlier image mockup. It included a dark jungle/gaming theme, top navigation, large hero title, green call-to-action buttons, feature cards, gameplay preview panel, and browser call-to-action section.

What worked:

- The website looked more professional.
- The design became closer to the required mockup.
- The page looked like an actual game landing page.

What did not work well:

- A leaderboard section was included but was later removed because it was not needed.
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Prompt 7 — Fixing Wrap-Around UI and Collision Rules

Prompt used:

The game UI is disturbed when the snake goes outside and comes from other side. Also, there should be a shake effect when the snake bumps to the wrong snake. A snake will eat from the mouth side. If large or poisonous snake

bumps or eat the player, game ends. Small snake should not bump the player itself.

Result:

ChatGPT fixed the wrap-around visual issue. The snake body was split correctly when crossing screen boundaries, so the canvas no longer showed a long broken line across the game area.

A shake effect was also added when the player hit the wrong snake.

What worked:

- The wrap-around bug was fixed.
- The shake effect made collisions feel more realistic.
- Dangerous snakes could now end the game by bumping into the player.
- Small snakes were made to avoid bumping into the player.

What did not work well:

- The rule about eating smaller snakes needed one more adjustment.
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Prompt 8 — Updating Small Snake Eating Rule

Prompt used:

Small snake should avoid itself to bump but the player can eat the small snake from anywhere.

Result:

The rule was updated. Small snakes still try to avoid bumping into the player, but the player can now eat a smaller snake by touching any part of that snake.

What worked:

- The game became easier and more enjoyable.
 - The rule became clearer.
 - Small snakes were useful as targets instead of being too difficult to eat.
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Prompt 9 — Removing Leaderboard

Prompt used:

Remove leaderboard.

Result:

The leaderboard section was removed from the landing page preview.

What worked:

- The landing page became cleaner.
 - The design became simpler and more focused on the game itself.
 - Unnecessary UI was removed.
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3. Final Game Features

The final version of the game includes the following features:

- A professional landing page
- Dark jungle-style game theme
- Hero section with game title
- Feature cards
- Gameplay preview section
- Playable game section
- Snake movement using keyboard
- Mobile controls and swipe support
- Screen wrap-around movement
- Food collection
- Increasing score
- Increasing levels and difficulty
- Player snake growth
- Enemy snakes
- Small snakes that can be eaten
- Big snakes that are dangerous
- Poison snakes that are dangerous
- Shake effect on wrong collision

- High score saved in browser
 - GitHub Pages deployment
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4. What Worked Well

Using ChatGPT worked well for generating the basic structure of the game quickly. It was useful for creating the HTML, CSS, and JavaScript files and for improving the project step by step.

The most useful part was the iterative improvement process. Instead of trying to create the final game in one prompt, I improved the game gradually by testing it, finding problems, and then asking ChatGPT to fix or improve those specific issues.

The best improvements were:

- Better game logic
 - Improved enemy snake behavior
 - Realistic snake drawing
 - Landing page design
 - GitHub Pages deployment guidance
 - Bug fixing through repeated prompts
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5. What Did Not Work Well

The first version of the game was too basic. The enemy snakes were not very realistic, and some game mechanics were not fully correct.

Another issue was that when the snake wrapped around the screen, the visual drawing became disturbed. This happened because the curved snake body was being drawn as one continuous line, even when part of the snake moved from one side of the screen to the other. This was later fixed by splitting the snake body drawing at wrap boundaries.

The landing page also required several iterations. At first, an image mockup was generated, but that did not update the actual code. The prompt had to be improved to clearly say that the website code needed to be updated, not just the design image.

6. How Prompts Were Improved Over Time

At the start, the prompts were general. For example, the first prompt only described the overall game idea.

Later prompts became more specific. Instead of saying “make it better,” I described exact problems and expected behavior.

For example:

- Instead of only asking for a good design, I asked for a landing page similar to the earlier mockup.
- Instead of saying the game was broken, I explained that the UI was disturbed when the snake wrapped around the screen.
- Instead of saying collision was wrong, I explained the exact rule: small snakes should avoid the player, but the player can eat small snakes from anywhere.
- Instead of keeping all generated features, I asked to remove the leaderboard because it was not needed.

This improved the output because the AI received clearer instructions and could make more accurate changes.

7. Challenges Faced

The main challenges during development were:

1. GitHub Deployment

At first, it was unclear whether ChatGPT could directly push the project to GitHub. The solution was to download the generated zip file, extract it, upload the files to GitHub manually, and then enable GitHub Pages.

2. Game Logic

The collision rules were more complex than a normal snake game. The game needed to handle food, small snakes, big snakes, poison snakes, player movement, enemy movement, and screen wrapping.

3. Snake Wrap-Around Drawing

When the snake crossed the screen boundary, the curved body drawing became visually broken. This required a specific fix in the drawing logic.

4. Realistic Snake Appearance

Making the snakes look more realistic in a browser game was challenging because the game used canvas drawing instead of image assets. The final version used curved paths, scale highlights, eyes, and forked tongues to make the snakes look more natural.

5. Iterative Prompting

The game could not be completed perfectly in one prompt. It required multiple prompt iterations, testing, and improvement.

8. Final Reflection

This project showed that AI tools can be very helpful for creating a working game, especially when the user provides clear and specific prompts. ChatGPT was useful not only for generating code but also for debugging, improving the design, and refining the game rules.

The most important lesson was that prompt quality matters. General prompts produce general results, while specific prompts produce better and more accurate results. The final game became much better after multiple iterations and clear feedback.

Overall, this project was a successful example of using AI to build and improve a web-based game through iterative development.

9. Conclusion

The final project, **Snake Survival Arena**, is a browser-based Snake Game created with the help of ChatGPT. It includes a professional landing page, playable game mechanics, increasing difficulty, enemy snakes, screen wrapping, and improved visual effects.

The game was deployed using GitHub Pages and can be accessed using the following link:

<https://aisha-batool.github.io/ai-prompting-asg1/>